Homework 11

① a) MDST is in NP

A problem is in NP if, given a potential solution (a certificate), we can verify in polynomial time whether the certificate is a valid solution.

Certificate: The set of edges $E_T \subseteq E$ that form the spanning tree T.

Certifier: we check three conditions in polynomial time:

1. Is T a tree?: Verify that (V, E_T) is connected and has $|V|-1$ edges.
2. Is T a spanning tree? If it is a tree on vertex set V, it is automatically a spanning tree.
3. Is the maximum degree of T at most k? Calculate the degree of every node in T and verify $\Delta(T) \leq k$. This takes $O(|V| + |E|)$.

Since all checks are polynomial, the MDST problem is in NP.

b) ~~[illegible]~~ MDST is NP-Hard

We reduce from Hamiltonian Cycle (HC).

Reduction:

Given an instance $G = (V, E)$ of HC, construct an MDST instance:

- Set $G' = G$
  so $V' = V$ and $E' = E$.
- Set the maximum allowed degree $k = 2$.

Proof:

($\Rightarrow$) If G has a Hamiltonian cycle.

- A Hamiltonian cycle contains all vertices exactly once.
- Removing one edge gives a Hamiltonian path, which is:
  connected, spans all vertices, has max degree ≤ 2
- Thus G' has a spanning tree of max degree 2.

($\Leftarrow$) If G' has a spanning tree with max degree ≤ 2

- Any spanning tree whose degree are ≤ 2 is a path.
- This path includes all vertices.
- If the endpoints are adjacent in G, adding that edge yields a Hamiltonian cycle.
- Thus G has a Hamiltonian Cycle.

$\therefore$ MDST is NP-Hard.

(2) a) k-Cycle-Decomposition is in NP

Certificate: A list of edge-sets $E_{C_1}, \dots E_{C_k}$, one for each intended cycle.

Verification (polynomial time):

1. Each C_i is a cycle of size a_i: Check that (V, E_{C_i}) forms exactly one simple cycle of length a_i.
   - $|E_{C_i}| = a_i$
   - exactly a_i vertices appear, each of degree 2
2. Cycles are vertex disjoint and cover all vertices: check that the vertex sets V_{C_i} are pairwise disjoint. Since $\sum_{i=1}^{k} a_i = |V|$, disjointness implies their union is all of V.
3. Edges are valid: verify each $E_{C_i} \subseteq E$

All verification steps run in polynomial time → k-cycle-decomposition is in NP.

b) k-cycle-Decomposition is NP-Hard.

Reduce from Hamiltonian Cycle (HC).

Given an instance $G = (V, E)$ of HC, construct the instance of k-cycle-Decomposition:

- $G' = G$
- $k = 1$
- required cycle lengths : $A = \{|V|\}$.

Proof of correctness:

If G has a Hamiltonian cycle, that cycle is a single cycle of length $|V|$. Thus G' has a 1-cycle decomposition of size $|V|$.

If G' has a 1-cycle decomposition of size $|V|$, the G' contains a simple cycle visiting all vertices exactly once i.e, a Hamiltonian cycle in G.

Therefore:
$HC \leq_P$ k-cycle-Decomposition

k-cycle-Decomposition is NP-Hard.

③ a) HALF-IS is in NP

- Certificate is a set $S \subseteq V$.
  We can check in polynomial time that:
  - $|S| = |V|/2$.
  - No two vertices in S are adjacent.

$\therefore$ HALF-IS is in NP.

b) HALF-IS is in NP-Hard

Reduce the NP-complete INDEPENDENT SET problem to HALF-IS:

Input (IS): A graph $G = (V, E)$ with $|V| = n$, integer k.
Question: Does G have an independant set of size at least k?

Construct in polynomial time a graph G' such that:

G has an IS of size k $\Leftrightarrow$ G' has an IS of size $|V'|/2$.

* Case distinction to simplify the reduction:

- Reduction assumes $k \le n$ (always true).
- We ensure $k \ge n/2$ by padding the original graph with isolated vertices if needed:

↳ if $k < n/2$, add $n-2k$ isolated vertices to G.
↳ Let the resulting graph still be called G.
↳ New graph has size $n' = 2k$, and the IS target is still $k = n'/2$.

We may assume that the input graph has size $n = 2k$.
This padding is polynomial.

<u>Reduction (assuming $n = 2k$)</u>

Define: $G' = G$
thus $|V'| = n = 2k$, $\frac{|V'|}{2} = k$.

HALF-IS on G' asks whether G' has an independent set of size k.

<u>Proof of Correctness</u>

($\Rightarrow$) If G' has an independent set of size k, then $G' = G$ has the same independent set of size $\frac{|V'|}{2} = k$.

so G' is a YES instance of HALF-IS

($\Leftarrow$) If G' has an independent set of size $\frac{|V'|}{2} = k$, then since $G' = G$, the same set is an IS of size k in G.

so G is a YES instance of IS.

$\therefore$ IS $\leq_p$ HALF-IS

so HALF-IS is NP-Hard.